5 Statistics

What students need to learn:

Ref	Content descriptor	Concepts and skills
SP a	Understand and use statistical problem solving process/handling data cycle	• Specify the problem and plan • Decide what data to collect and what statistical analysis is needed • Collect data from a variety of suitable primary and secondary sources • Use suitable data collection techniques • Process and represent the data • Interpret and discuss the data
SP b	Identify possible sources of bias	• Understand how sources of data may be biased
SP c	Design an experiment or survey	• Identify which primary data they need to collect and in what format, including grouped data • Consider fairness • Understand sample and population • Design a question for a questionnaire • Criticise questions for a questionnaire
SP d	Design data-collection sheets distinguishing between different types of data	• Design and use data-collection sheets for grouped, discrete and continuous data • Collect data using various methods • Sort, classify and tabulate data and discrete or continuous quantitative data • Group discrete and continuous data into class intervals of equal width
SP e	Extract data from printed tables and lists	• Extract data from lists and tables
SP f	Design and use two-way tables for discrete and grouped data	• Design and use two-way tables for discrete and grouped data • Use information provided to complete a two-way table

Foundation

Ref	Content descriptor	Concepts and skills
SP g	Produce charts and diagrams for various data types	• Produce – Pictograms – Composite bar charts – Comparative and dual bar charts – Pie charts – Histograms with equal class intervals – Frequency diagrams for grouped discrete data – Line graphs – Scatter graphs – Frequency polygons for grouped data – Ordered stem and leaf diagrams
SP h	Calculate median, mean, range, mode and modal class	• Calculate: – mean – mode – median – range – modal class – interval containing the median • Estimate the mean of grouped data using the mid-interval value • Find the median for large data sets with grouped data • Estimate the mean for large data sets with grouped data • Understand that the expression 'estimate' will be used where appropriate, when finding the mean of grouped data using mid-interval values (NB: Quartiles and interquartile range are Higher Tier only)

Ref	Content descriptor	Concepts and skills
SP i	Interpret a wide range of graphs and diagrams and draw conclusions	• Interpret: – composite bar charts – comparative and dual bar charts – pie charts – stem and leaf diagrams – scatter graphs – frequency polygons • Recognise simple patterns, characteristics and relationships in bar charts, line graphs and frequency polygons • From pictograms, bar charts, line graphs, frequency polygons, frequency diagrams and histograms with equal class intervals: – read off frequency values – calculate total population – find greatest and least values • From pie charts – find the total frequency – find the size of each category • Find the range, mode, median and greatest and least values from stem and leaf diagrams
SP j	Look at data to find patterns and exceptions	• Present findings from databases, tables and charts • Look at data to find patterns and exceptions
SP k	Recognise correlation and draw and/or use lines of best fit by eye, understanding what these represent	• Draw lines of best fit by eye, understanding what these represent • Distinguish between positive, negative and zero correlation using lines of best fit • Use a line of best fit to predict values of one variable given values of the other variable • Interpret scatter graphs in terms of the relationship between two variables • Interpret correlation in terms of the problem • Understand that correlation does not imply causality

Foundation

Ref	Content descriptor	Concepts and skills
SP l	Compare distributions and make inferences	• Compare the mean and range of two distributions • Understand that the frequency represented by corresponding sectors in two pie charts is dependent upon the total populations represented by each of the pie charts • Use dual or comparative bar charts to compare distributions • Recognise the advantages and disadvantages between measures of average
SP u	Use calculators efficiently and effectively, including statistical functions	• Calculate the mean of a small data set, using the appropriate key on a scientific calculator